

Mateo Alado

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EDUCATION

California State University, East Bay

August 2022 – June 2026

Bachelor of Science in Computer Science

Hayward, CA

- Distinctions: 2x AI Research Publications, HackHayward 2026 Winner, 3x Dean's List
- Coursework: Data Structures and Algorithms, Software Engineering, Operating Systems, AI, Statistics

PROFESSIONAL EXPERIENCE

Software Engineer Intern

June 2026 – August 2026

Lawrence Berkeley National Laboratory

Berkeley, CA

- Engineered a pipeline of 4 LangGraph agents for scientific literature ingestion and retrieval of 16,000+ publications
- Built a PostgreSQL-GraphQL API for data retrieval, cutting task completion time from 3+ minutes to <5 seconds
- Deployed agents on an HPC cluster using Slurm, scaling literature extraction and processing throughput by 70%
- Created 100+ end-to-end backend/agent tests with pytest, increasing total codebase test coverage 95%+
- Containerized deployment of agents and services with Docker, expanding access to 450+ ALS beamline scientists
- Resolved a request concurrency bottleneck by implementing a semaphore, increasing concurrent API calls by 25%

Software Research Engineer Intern

January 2026 – May 2026

California State University, East Bay

Hayward, CA

- Developed a FastAPI backend with context retrieval for data management, cutting request context size by 79.7%
- Built a script for automating data migration to a NoSQL database via cronjob, improving processing time by 60%
- Optimized an adaptive BFS graph traversal algorithm for retrieval, improving Neo4j response relevance to 89%
- First-authored an IEEE paper on GenAI applications in academic research for the GAISS 2026 conference

Software Engineer

August 2025 – April 2026

The San Francisco Tooling Company

Remote

- Built a TypeScript agent harness with 70+ tools enabling CI/CD management, PR reviews, and QA automation
- Built an authentication service with Auth0 and Vercel, enabling secure credential management for 20+ engineers
- Migrated MCP tool execution to AWS Lambda for serverless hosting, reducing overall service latency by 65%
- Designed a React UI dashboard with real-time API and tool usage statistics for users, increasing observability

PROJECTS

Perplx - HackHayward 2026 Winner | *TypeScript, Bun, Cloudflare Workers, Cloudflare D1, MCP*

- Developed an AI coding agent with MCP tooling using the Perplexity API for context-aware software engineering
- Reduced cost per completed task to approximately \$0.10 through auto compaction and prompt caching
- Built a session-sharing infrastructure using Cloudflare Workers and D1 enabling real-time collaborative sessions via shared URLs for 100+ concurrent users

Publication Management Platform | *Python, TypeScript, React, FastAPI, Neo4j, Supabase*

- Built and deployed a full-stack RAG research platform using React and FastAPI for 20+ users to query and manage 100+ research publications and datasets
- Implemented persistent session context retrieval and user data storage with Supabase and Neo4j, cutting end-to-end latency by 75%+
- Integrated LiteLLM to benchmark model accuracy and performance for 5 models across 3 different LLM providers

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C, C++, Go

AI/ML: LLMs, AI Agents, LangGraph, LangChain, RAG, PyTorch

Backend: FastAPI, REST APIs, GraphQL, Node.js, Express

Data: PostgreSQL, MongoDB, NoSQL, MySQL, Supabase, Neo4j

Frontend: React, React Native, Next.js, Tailwind

Cloud and Tools: Docker, AWS (Lambda, DynamoDB) Git, GitHub Actions, Linux, UNIX, Cursor, Claude Code

Concepts: Backend Development, Fullstack Development, Cloud Infrastructure, GenAI, CI/CD, DevOps, Agile